University Of Engineering and Technology, Lahore
Computer Engineering Department

Lab Evaluation:

Course Name: Data Structures & Algorithm	**Course Code:** CMPE-212L
Assignment Type: Lab	**Dated:** 23rd April 2024
Semester: 4th	**Session:** 2022
Lab/Project/Assignment #: 11	**CLOs to be covered:** CLO 2
Lab Title: AVL Tree	**Teacher Name:** Engr. Afeef Obaid

CLO 2	Design solutions for problems using appropriate data structures and algorithms.					
Levels (Marks)	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
(10)						
					Total	**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	**9-10**	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	**7-8**	L2	Submitted the lab tasks but have good understanding
Good	**5-6**	L3	Submitted the lab tasks but have weak understanding.
Basic	**3-4**	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	**1-2**	L5	Submitted only one lab task.
Not Acceptable	**0**	L6	Did not attempt

LAB No. 11

Lab Goals/Objectives:

- Insertion of node in AVL Tree
- Deletion of node in AVL Tree

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Equipment Required: Computer and Dev C++ Compiler

Theory/Relevant Material: AVL Trees

Lab Tasks:

- Complete the definition of the following functions of AVL Trees in the code provided. Implement the following functions of AVL
- Insert node in AVL Tree
- Height of Sub Tree
- Balance Factor of any node
- Rotation of Nodes in AVL Trees (All Cases)
- Deletion of nodes in AVL Trees